Name	ID	Department
Abdelrhman Mohamed Abdelhady Hodib	2022513643	Intelligent system

Defining the process and cpu scheduler classes

In [1]:
```
class Process:
    def __init__(self, pid, arrival_time, burst_time):
        self.pid = pid  # Process ID
        self.arrival_time = arrival_time  # Arrival time of the process
        self.burst_time = burst_time  # Burst time of the process
        self.start_time = None  # Start time of the process
        self.finish_time = None  # Finish time of the process
        self.waiting_time = None  # Waiting time of the process
        self.remaining_time = burst_time  # Remaining time of the process

class CPU_Scheduler:
    def __init__(self):
        self.queue = []  # Initialize an empty list to represent the queue of processes
        self.current_time = 0  # Initialize the current time to 0

    def add_process(self, process):
        self.queue.append(process)  # Add the process to the end of the queue

    def run(self):
        self.queue.sort(key=lambda x: x.burst_time)  # Sort the processes in the queue by burst time
        for process in self.queue:
            waiting_time = max(self.current_time - process.arrival_time, 0)  # Calculate the waiting time for the process
            self.current_time += process.burst_time  # Update the current time to account for the process's burst time
            process.turnaround_time = self.current_time - process.arrival_time  # Calculate the turnaround time for the process
            process.waiting_time = waiting_time  # Set the waiting time for the process

    def get_avg_waiting_time(self):
        total_waiting_time = sum(process.waiting_time for process in self.queue)  # Calculate the total waiting time for all processes in the queue
        return float(total_waiting_time) / len(self.queue)  # Return the average waiting time for all processes in the queue
```

Applying the algorithm

Process	Arrival time	Burst Time
P1	0	6
P2	1	4
P3	2	2
P4	3	3

In [2]:
```
# Create some processes with process ID, arrival time, and burst time
p1 = Process(1, 0, 6)
p2 = Process(2, 1, 4)
p3 = Process(3, 2, 2)
p4 = Process(4, 3, 3)

# Create an instance of the SJF CPU scheduler
scheduler = CPU_Scheduler()

# Add the processes to the scheduler's queue
scheduler.add_process(p1)
scheduler.add_process(p2)
scheduler.add_process(p3)
scheduler.add_process(p4)

# Run the scheduler
scheduler.run()
```

```python
# Get the average waiting time for all the processes in the queue
print(f"Average waiting time: {scheduler.get_avg_waiting_time()}")
```

```
Average waiting time: 3.25
```